# RAPPORT DE SOUTENANCE

Lossless inc.

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# I) The Story

During the first meeting of our group, we decided the story we wanted to represent in our game. We wanted to study the path of grief, for a little girl growing up experiencing that feeling. We find ourselves brought into the life of a small family, a father and a daughter, living in a small house in the forest. After rushing to see her dad, the young girl, Belladona, bumps into a candle, which becomes the turning point of the story. The entire house burns very rapidly, trapping inside the dad, who managed to get his daughter out of the place in a short time. Ordered by her father, she ran through the forest to find shelter, but encountered a new character, a lost heart. Filled with sadness, fear, and remorse of leaving her father, Belladona decides that this heart will be her companion during her travels. After some time, both decided to stop at a river to rest, but as they were bending over to reach the water, Belladona fell into it. Controlled by the current, she couldn't stop and realized that she was going to plunge into a waterfall. Without managing to carry herself out of the water, she began her downfall, passing out during her drop. When she finally wakes up, she is lying on the ground, soaked in water, the heart sitting next to her. From that point, Belladona must make her own path to reach the end of her journey, hopefully a better place. She walks through seven worlds, the seven stages of grief as we know. For each of these worlds, she is going to have to fight her own enemies: the villains that only want to retain her from growing, evolving, and moving on. Both characters follow the river up and will be confronted to many tedious and bizarre adventures.

# II) Eden's part

## A) Graphics

**a. First step of the elaboration of the graphics of our game: the mood board (cf appendices 1 to 5)**

On the first meeting of our group, we took some time to decide first what ideas we wanted to research for the main story, but from that point, when we knew we were willing to study our character's path during grief, we decided to determine the colorimetry of the game. After some exploration on the web and on Pinterest to find inspiration for the colours, I created a board (cf appendices 1 and 2) with multiple sub-board inside of it for the different stages of grief. Each photo that I put in my J2D board represents the main idea I wanted to convey in the elaboration of the game. The mood board then only gathered the pictures I put into these sub-boards, so that everyone could know what the atmosphere I wanted to represent was. I put multiple pictures together and added some descriptions and details I wanted to precise with the people of the team. The atmosphere of each period of the game changes quite drastically, since as the steps of grief are surmounted, the action takes place in a completely different location, with different colours and characters.

**b. The beginning of the game: the flashbacks (cf appendices 6 to 10)**

For the beginning of the game, we gathered with everyone to decide about the most important details for this part, the start of the story, as it is at that moment that the player and the tester will be completely seized by the drawings but also the story. Amelie and I did a second smaller mood board and timeline of how we wanted to represent that part, to be the most precise possible (cf appendix n°6).

From the moment the mood board and the timeline was validated by everyone in the team, I started working on the way to represent the story, in the best manner possible.

I started by the beginning of our story and of our game, as to introduce the game to the future player but also to myself, so that I could really understand the challenges of it. To draw these, I used a drawing application, Procreate, that cost me 15€ (cf appendix n°11), but as it is the best one on the market, it was not a hard decision to make. As the story begins with a

small family living in the forest, I represented that as my first flashback. The rest flowed, as the narrative was very clear. In total, we have around 10 images that will work as animation for the introduction.

### c. The backgrounds of the game

After the end of the introduction, the game begins, and there appears the first world of our game. In terms of technical aspects, each world is divided into sub-worlds. Within each world, 3 rooms will be included in a randomized order. So, when our character finishes the first room of the world, another one follows, without knowing which one it is. For the graphical part, as our story focuses on the different stages of grief, here denial & chock, anger & bargaining, sadness, acceptance, resignation, and finish with reconstruction, we represent each of these steps in the following way. The first world is denial and chock, and it takes place in an inhabited village inside a forest. I represented the colours of this theme in black and white, with different versions of grey. In order to reach the perspective, I wanted to represent, I drew in these different shades of grey, beginning with black, then dark grey, to in the very most background reach a very light grey (cf appendix n°12). This perspective method works with the different layers that I created when drawing. In total for the first world, I used 27 layers, mixing these different colours but also the constructions I did. By layer, I did 3 different options of disposal, as we want 3 rooms inside this world. The same method is replicated on every other one, but the atmosphere changes drastically from one to another. The second world being anger and bargaining, the colours evolve around shades of red (cf appendix n°11). Anger is a feeling of explosion and hard thinking, so I decided to make the character walk through a world of destroyed buildings and deserted cities. The third world is sadness, a sentiment of loneliness and coldness. Our protagonist will be surrounded by glass and snow, but for this one, the rooms will not be in a randomized order, as the second one will take place in a cave, and I want to represent a continuous path, so the surrounding rooms will show the entrance and the way out of the cave. The randomization however continues in the next world, the acceptance phase. In this one, the colours will be evolving around a light purple one. The atmosphere will be quite luminous, as the character has no choice to accept and continue her path towards the end. The resignation phase shows a lot of strongness of mind but also a lot of determination. It is the last phase before reconstruction; it has an impact on our character. For this one, the atmosphere will be progressing around a dark purple colour. She is affirming herself in her decisions and has no other will but to reach her growth. The last and final world is the reconstruction chapter. The light will be exploding, we wanted to represent a cyber-punk ambiance, a mix of every feeling our character has been living from the beginning. She will radiate through the place, ready to fight the final boss of her journey. The graphic part is very important, as it is the main source of atmosphere the player will feel. It is why I focus on the good realisation of this task.

## B) Website

### a. Our domain name

To have a proper domain name that affiliates to the site, we bought one for 25€, on WIX. The domain name is called losslessinc.fr, we wanted to have a website that fully represents our game, so it is better to name it after our group name. The process of changing our link name is not yet done, it will be for the next defence.

### b. Composition of the website (cf appendices 13 to 17)

As expected by the guidelines, the website divides itself into multiple parts (cf appendix n°13): the presentation of our game, a second part with the images of it, the next a presentation of our group, then all the references used to create the game, and lastly a part where you can download report of each previous defence. For now, I only focused on the technical part, so the html coding, the CSS will be done later in the next weeks. For each of these pages, I filled them with the text and some images we wanted to add inside:

- In the presentation of our game, I wrote the story of the game, and I will add the different steps our characters will go through, and the bosses she is going to have to fight on her way.
- The next one is the presentation with images. This time, I took the images of some flashbacks and the menu part to fill it, we are going to create some more especially for the website.
- The following page is the presentation of our group. I added a small text that gathers some explanations about the different people of our team, and I will add some pictures of ourselves. We will talk about who we are, our passions, and how it led to the creation of our game.
- The “all used” part will be the page that explains which references were used during the making of the game, so the books and articles employed.
- Lastly, with the page “downloads”, you can get all the 20 pages and more that we are going to do for all the defences.

At the moment, the structure of the site is done, we only have to make it appealing. The code for it is divided into the same number of pages, and one more that gathers the links to all of the different sub-sites. The index is the opening of the website, while the other ones

gather the texts of each page. To include the pictures inside of them, I created a file that stocks all of them, then I link them in the code. For the second defence, the appealing part will be done, at least half of the website will be complete.

## C) Scripts

### a. The beginning of the game: the flashbacks

During the beginning of the game, as explained earlier, a sort of animation will take place with a sequence of images. For each of these flashbacks, a small text will appear on the screen, for the player to fully understand the story before it plays. As each flashback is already quite precise about what's happening, the texts didn't need a lot of details, they just needed to introduce the story. The first flashback (cf appendix n°7) shows the character and her dad on a river, the text explains that a bit more: *"Like every sunny Sunday Belladonna and her father had, they went fishing on the lake next door to their house. It was a peaceful ritual, a kind of tradition between them since Belladonna's mother passed away."*. For the second one, we are transported inside the house: *"After their successful day, they deserved some rest. While rushing to go talk to her dad about a meaningless topic, as usual, Belladonna bumped into the cupboard, leading to a candle falling on the ground."*. As the next one is only a close up on the candle, no script was needed, but the next one is a drawing of the house burning, so the text precise this detail: *"The house burned very rapidly, as it was entirely made of wood."*. When our character is seen running outside of the house, this is what is written: *"Everything burned, but Belladonna's dad managed to get her out of the place on very short notice but unfortunately trapped himself inside of the house."*. When she reached the river shown in the picture: *"Her father ordered her to run away far from the flames to find shelter, and so she did. She ran, ran and ran, until reaching a river that would allow her to take some rest."*. One before last, it's the description of the beginning of the climax: *"As she leaned forward on the water, she fell into it. But the current was too strong, and too rapid, so she didn't manage to get herself out of the water in time. She began her plunge into the waterfall but didn't realize as she passed out during her fall."*. Lastly, the flashbacks come to an end with: *"She woke up shaken by a surreal character: a red heart, his face marked with worry. The heart explains to her that he found her after some time, as he deduced by her soaked clothes and wet hair."*. Then the game begins, and no script this detail will appear until the end of the game, depending on the good or bad ending the player chooses. This chapter is not yet done, but I wrote some other parts.

### b. During the game: the merchants

While the two characters continue their adventure in the game, they are going to encounter other talking figures, here we focus on the merchants. These characters will help Belladonna and the heart during her travels, especially for the weapons and objects they can buy. These characters are very common in every roguelike game, so their possible words could be: *"Hello Belladonna/Heart, I'm _____ what would you like to buy?"* if this is the first visit to this merchant, but for the next ones : *"Always glad to see you" "What do you need today ?" "How can I be of any assistance today?" "Can be of any help?"* for the beginning of the discussion, and for the end of it, if she buys a weapon: *"Thank you!" "Be sure to come again!" "See you next time!" "Don't hesitate to come back!"*, and if she doesn't buy anything: *"I'll be there next time!" "No rush!" "That's fine for me!" "Always happy to see you!"*. These interactions contribute to the installation of a good mood between the player, Belladonna, the Heart, and the Merchant, so that the game feels reassuring when she doesn't fight the bosses.

**c. During the game: the traveling fish (cf appendix n°21)**

This character helps the two players to go from one point to another. The travel is possible once Belladonna and the Heart have already discovered the world, so they can only come back to where they have been. From that, if it's the first time that they travel with the fish, it might say: *"Come travel with me!" "I can get you to every point you want!"* or *"I'm here to assist you in any transportation kind."*. If the characters have already travelled with it, the opening of the discussion might be: *"Where to this time?" "Once again I am here" "Don't hesitate to travel with me"*. The fish will be a good help for the two characters, and as its presence is a necessity, his speech must be welcoming for both Belladonna, the heart and the players.

## D) End music

**Piano part**

The ending music with the generic will have a special soundtrack played by me on the piano, the 3rd impromptu by Schubert, a piece that conveys peace and calmness. For both the good and the bad ending this soundtrack will take place, as it doesn't rely on any story element, except for the reconstruction of Belladonna and her growth through grief.

## E) Documents

**a. CDSF**

For the Oral in class, we already had to prepare a big part of this presentation, so everyone contributed to do their part. As our teacher had some recommendations, I added the missing details to our text. Every element was validated by everyone on the team.

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**b. CDST**

Except for the personal parts of this presentation, I answered the questions and assembled everything. The presentation will be using the same graphics that Amelie has done for the CDSF.

# III) Amélie’s part

## A) Graphism

### a. The conception of the game’s art

I am going to start by listing all the things I actively participated in for our group project. We started the making of the game with a simple idea, making a game centered around grief. As the lead graphic designer of our team, making the graphics team work together to make an artistically coherent game was one of my main priorities. As such, one of the first steps in our project was to design and create concept art about the game, its environment, characters, designs as well as creating mood boards that would capture the idea of the game we wanted to convey. This was particularly important since the graphics team is composed of three people, we need to have our ideas mapped out so everyone could be aware of the art direction of the game and follow it closely.

### b. The conception of the main characters

At first, we all created our own vision of the design of the main characters and ended up settling on the current design (cf. Appendix n°18 and 19), I designed the character sheet of the girl (cf. Appendix n°19). We have the two main characters, the one played by the player 1, Belladonna, a young girl who just lost her father to a housefire and in fleeing said housefire ended up in the strange forest the game takes place in. The other main character is a Heart, he is played by the Player 2 and represents another way for grief to manifest through a positive façade as well as the emotions of Belladonna and its manifestation which in our game she would have to learn to accept. We also settled on using a pixel-art art style for everything in the game.

### c. The flashbacks

I then worked with Eden on the flashback that would be shown at the start of the game; to give the player some exposition, this thus led us to discuss the story in more detail. Eden would work on the background, and I would add the characters and the light on top of the drawings. We started by making a storyboard of the flashbacks for the character and player 1 Belladonna (cf. Appendix n°6), it starts with her living a happy life with her father and due to a sudden fire burning down the house, she was forced to flee

her house leaving her father in the process, and ended up falling down a cascade. The first panel is Belladonna and her father on a boat in a lake with trees and bushes around them next to her house, the colours are bright and light, next they are inside the house, a candle falls on the ground, the colours are still bright but they are warmer on the red side, to show the contrast with the outside as well as a transition towards the fire that happens in the later panels. Next, the house on fire is supposed to contrast with the green and lush forest. Belladonna will then rush towards the forest, and end up even more lost, at this point the colours are dull and cold, it is the same forest she knew her whole life, yet it is unfamiliar and hostile towards her, this leads her to fall from the cascade. The flashback is supposed to give the player context without giving away the whole story, it gives the player and the character a goal; to return or at least to find a home, away from the strange forest. It is a metaphor for her grief and loss, even things that brought her joy before, the water and the forest end up unfamiliar and strange for Belladonna. We have yet to finish the concept and flashback for player 2.

### d. The importance of the river

I also made some concept art for the gameplay and menu screen (cf. Appendix n°20). For the pause screen I got inspired by the painting and the tale of Ophelia from Hamlet, mostly the painting by John Everett Millais and the poem Ophelia by Rimbaud, where Ophelia lets herself float on a river amidst nature, floating towards her death due to a broken heart and madness. This gave me the idea of making the game follow the same river the whole way through as a symbolic metaphor for the state of mind of the first main character Belladonna, who is experiencing grief for the first time in the form of the loss of her father. As such the river became more than just a simple background element and ended up as a perfect metaphor for the grief and loss Belladonna experienced. Therefore, the river became an important, I would say even central part of our project where a lot of the art and concept shifted toward this river.

### e. The river's inhabitants and the worlds

For example, a lot of the passive or friendly non-playable characters (NPC), are centered around river creatures, such as the Travelling fish for which I have made concept art (cf. Appendix n°21), this fish would serve as a form of exposition for the character as well as a way to travel between the five worlds the game is centered around, which represents the five stages of grief. The design ideas this fish was centered around was a river fish that could easily travel through the muddy water of the river while being big enough to support people travelling on its back. In the end I settled for a Catfish, they are quite big once they have reached their adult size and live on the bottom of the river floor. They are also known for their endurance and resilience therefore perfect for serving as transportation through the muddy water they live in. Another creature we created is the Oyster lady who got lost and ended up floating into the river. She is a fortune teller, who

helps the player acquire skills through sacrificing cards they found on their adventure (The card system is explained later in the to be done part). There is also the merchant toad, which would sell and hold onto weapons for the players.

Although there are five worlds the player fight in, the actual game is composed of 7 different places, they are placed in the flow of the river which is in a valley (cf. Appendix n°22), the denial world is placed at the top of the first hill, then the river flows down towards the valley down to the sadness crystal cave and goes back up another hill to reconstruction. The first area is the Merchant's Creek (cf. Appendix n°23), before the denial world, serves as a hub where the players can rest, talk with the local fishes and exchange goods. The creak is arranged with the river in mind first, it is constantly in the background, and we can also see the cascade Belladonna and the Heart fell from at the start of the game. It is supposed to be imposing and grand as the biggest and major part of the place. The creek is also a hub for the merchant fishes of the world to gather and exchange goods with the players. It's where we can find the merchant toad, the oyster lady and the travelling fish. So far these are the fishes that inhabit this place, but if needed we could add some other specialized fishes that would aid in the characters and player's journey.

After the merchants creak lies the five worlds representing the five stages of grief, and finally we have reconstruction which is where the final boss fight will happen. As per the scheme (cf. Appendix n°22), this place is on the top of the second hill and therefore towers over the rest of the game. This concept art of the final fight (cf. Appendix n°20) is supposed to be a symbolic representation of Belladonna and the Heart fighting and the grief itself to finally be free from it. The sun is setting down as it is the end of their journey through their emotional introspection. The light is bright and warm with the sun illuminating their path. It is the culmination of their journey and a test of what the players and characters have learnt about the game so far. Depending on the result of the fight there are two possible endings, either they lose and we have a cutscene where we show that they are at the start again, having to brave through the forest once again. Or they win and they find their way back home.

### f. The sprites

Tarek and I started working on the sprites of the characters, I was tasked with creating the sprites for the player 1 Belladonna (cf. Appendix n°24), I made an eight-frame running cycle as well as sprites when she isn't moving. I tried to capture the rush and determination she must get through this forest despite the terrible emotions she feels through the grief.

### g. The presentation

As a part of our presentation of the game, I made a slide show with the river representing the five primary stages of grief and therefore the five worlds the players will fight in.

## B) The Menus

I also worked on the conception and the making of the main menu of the game, its functionalities and art. The main menu is really important to the game as it is the player's first introduction to the game and its style, as well as its functionality.

### a. The conception of the menus

The menu would work as such: the game would start with a fade in onto the logo of our group lossless inc., and then it would fade onto the warnings of the game since it tackles topics such as depression and more. The game would then open up the main menu page, the background had to contain the two main characters, the river and the house, this was I think the most important elements, as we can see in the first draft I made of the background (cf. Appendix n°25). For the buttons, since this is a two-player game that works such that the player one holds the saves for both players and any player 2 can join onto the saves, we needed a button and a page that would access the saves and one that would access the way to join the player 1. The server works in local, therefore we needed a way for two computers to connect, and we ended up settling with using the IP address of the host (therefore the player 1) as a code the player 2 has to enter in the join page. We also have an option page, where we have yet to decide on what to incorporate. There is also an exit button to quit the game. This was the general idea I had for the main menu of the game.

### b. The making of the main menu

With Eden, we then made multiple drafts for the menu and ended up with the finished image (cf. Appendix n°26) with the main characters in the middle of the screen, the river in the background leading up to the house, which is behind the title card. I animated the menu making it look alive with the trees and the river moving in the background and leaves blowing on top of the characters. The page system works with a variable "action" that dictates which page the player is currently on. Every time we change pages from the main menu an animation I made where the characters fall in the river plays, this happens as a transition since in the other pages we see these characters in the water

of the river, again demonstrating the importance we give to the river. In the Join page (cf. Appendix n°27) (the page where the player two can join a session) there is an animation of the heart in the water since the heart is the character we play as if we are the player 2. The eyes of the heart follow the mouse of the player around the page; in the middle we have the buttons that allows the player to write the IP address of the host. In the Play page (cf. Appendix n°28), the transition for this page has a use, since it launches the server on the computer allowing for smooth transitions down the line. For the background, we have an animation of Belladonna in the river this time with her eyes following the mouse as well. There are 3 possible save files, the player can choose from, when choosing one it will trigger the waiting for player screen as well as show the IP address of the player on the screen for the player 2 to copy. In both pages, we see the characters in water as if they really fell into the river, it also serves as a transition for the player to get into the game. The option menu is slightly different (cf. Appendix n°29), we have an animation of the character, but the camera is inside of the river, for now we have yet to add anything to the option menu.

One other thing that is important for the menu but also the display of the game in general is the resolution. Since the art direction is in pixel art we settled on using canvases of 480 x 270 which is not nearly enough to fill the entire screen of a regular computer, therefore at the beginning of the program the entirety of the images and buttons are scaled up by a coefficient calculated with the current resolution of their screen. Having the game play at any resolution was very important for me as I think it raises the quality of life and experience of the game. What's more, with the help of Pierre finding sound effects on the internet, I was able to add them to the buttons of the menu, this is just a start on the many sound effects we plan on adding later on.

## C) What needs to be done

### a. The game loop and non-playable characters

Now I am going to explain the things that I still need to do for the project. I want to start working on a lot more sprites for the different characters, players and enemies of the game so we can really start implementing the game loop, with the different challenges of our game. As well as the art for the rest of the menus such as the pause menu and the ending screen.

The combat system also needs to be polished, the card system I mentioned is one of them. Both players possess their own collection of cards that work like skills in the actual combat. We can carry only one card per world. For a bit of challenge, the player 1 is the only one that can gain new cards though defeating bosses. The only way for the player 2 to gain new cards is for the player 1 to sacrifice some of his to the oyster lady which would give the player 2 a new card.  As such this is a major part of the game and the combat system and is to be added in the near future.

**b. Screen resolution**

The option menus need also to be finished, we already have a list of all the different things we want to add in this page such as audio controls, exit, load and save buttons and maybe if possible, options that allows the player to change the resolution and the screen of the game. The resolution of the other aspects of the game such as the player, enemies and rooms also need to be tackled so it can fit inside any screen the player could have

# IV) Tarek's part

As a team member, most of my tasks rely on the visual and literary side of things.

## A) Graphics

### a) Deciding on our characters' visual identities

Before starting to design our characters; it was very important to know what direction we wanted them to take look wise. As a result, we wound up discussing how we could link our central themes of grief with the way the characters looked. We knew we wanted to convey their states of mind and personalities through simple and easily recognizable design elements. Thus, dawned upon us the use of shapes as a visual language of our own. For instance, our main character, Belladonna, is a young, emotionally closed off young girl who seeks refuge from the outer world in her own silence. Representing this dominant aspect of our character was fairly easy, as I first came up with the idea of her wearing a coat that would engulf her whole body, leaving only the upper side of her face, as well as her legs to show, as it figures in early concept art (cf. Appendix n.31) for our character. This idea was further expanded upon later, although it consisted in a change of aesthetic, making Belladonna seem much less edgy, appearing with much softer features. Her coat also changed, becoming more fitting of a cottagecore aesthetic.

### b) Sprites and animations

This was the next step towards starting to fill this world of ours with life. By collaborating with Amélie, our lead graphic designer, I was able to create sprites for our main supporting act, The Heart (cf. Appendix n.18). We wanted them to imitate a water droplet in shape, as our game is centered around the concept of a river of grief. As such, I made it so The Heart would undulate as they went by, mimicking the flow of water.

In a more concrete and technical direction, implementing our sprites required us to make sprite sheets. I personally made a 12-frame animation for The Heart, trying to mimic the movement of water, as said earlier. Furthermore, the animations were made using Procreate, an iPad art app. All of my frames were drawn by hand, at a 15-fps framerate, and animated "*on twos*", which is an industry term referring

to the act of doubling each frame of the animation so as to ensure it flows smoothly. (cf. Appendix n.32).

## B) Writing

### a) World Building

Aside from visual matters, I also actively contribute to our world building. Our game is centered around grief. As such, Belladonna and The Heart must go through various areas of the river, each symbolizing core emotions ranging from denial to reconstruction. In doing so, she must face bosses that represent each one of these emotions. For instance, throughout the narrative, Belladonna is led to go through a mysterious area we decided to name "*The Veiled Bend*", a desert, mysterious place which appears to have once been called "home" by a clan of guardian crocodiles. These days, travellers actively avoid this area, as it has now been turned into headquarters for the *Nutria Mob,* a collective of antagonistic, barbarian and invasive nutrias (cf. Appendix n.33). This choice of species was very much intentional on our part, as crocodiles are natural predators to nutrias, keeping them in check. As the river serves as an allegory for Belladonna's grief, the crocodiles were once the guardians of her emotional stability and peace. As her mind was deeply affected by her father's loss, the crocodiles found themselves to be overwhelmed by the invasive nutrias, symbolizing the paralyzing effect of loss on individuals.

### b) Character writing

Like I previously let on, I am also tasked with writing our project's characters, in collaboration with Eden. Characters like Belladonna are heavily marked by their emotions, whereas other characters, such as the mysterious Amri-Sa, aid her in growing into her feelings, by blessing her with their wisdom throughout her journey. Here are excerpts of both aforementioned characters :

***Belladonna*** :

- Struggling with emotions since a young age, she has always been perceived as a strange child, hiding some dark and intangible feelings inside her heart.
- Her struggle with emotions could come from the fact that the nature around her reacts to her feelings.
- Intense emotions imply equally intense phenomena; the world becomes a metaphor for her headspace.

***Amri-Sa*** :

- His name stands for "*the one who makes the heart quiet*" in an ancient, foreign language.
- He represents acceptance within Belladonna's heart. In a mysterious way, he mirrors her insecurities.

- As a reptilian, his cold-blooded nature often misguides people into thinking he is void of emotions: He's just at peace with himself
- This mysterious crocodile appears from the beginning of the game. He conceals himself as the riverbank merchant, always helping Belladonna. He could be the one to ensure her safety after the events that trigger the story.
- Could make a play on his name: After his fight, Belladonna's heart's tone changes. His character grows into its own: Fighting those stages of grief instead of making peace with them creates accumulating resentment/doubt/turmoil within it. " Why does our peace rely on my tolerance for your unhealed wounds?". The one who makes the heart quiet inspired great turmoil in it.
  The character relationships were intended to be intentional, and useful to advance our narrative. The Heart, for instance, is a vessel for each step of Belladonna's healing process, which allows us to tie this aspect of their character to broader story advancement and gameplay mechanics. It was also essential to use this narrative tension of "*fighting our emotions*" to move through our pain. The game asks a fundamental question: "*Should we fight or coexist with them?*". Once again, The Heart's story arc revolves around this question, as he is made to absorb every single one of our main character's emotions throughout this story. Will he build resentment, or behave like a simple, emotionally void narrative tool? One must Wonder.

# V) Pierre’s part

## A) Server and Client

### a. Research

During the first semester, I carried out a wide range of tasks whose primary objective was to establish a solid technical foundation for the rest of the development. This phase required a significant investment in research and experimentation (trial and error), as the server and client network I developed during this stage would directly influence the final game.

One of my earliest priorities was to understand how multiplayer video games operate from a networking perspective. To achieve this, I conducted extensive research using online documentation, technical articles, developer forums, and educational resources. Thanks to this research I’ve been able to understand how multiple players can be connected simultaneously within the same game environment while maintaining consistent game states. After studying different multiplayer architectures, including peer-to-peer and client-server models, I chose to settle on peer-to-peer connection, since it was the easiest connection method to understand, implement and debug.

However, not all of this research proved directly useful. Some tutorials and examples were either outdated, overly simplified, or designed for engines and languages different from our Python and Pygame environment. For example, I attempted to implement a networking solution that was intended for turn based-games, which resulted in a design that was inefficient for a fast-paced multiplayer gameplay. In this failure, I learned not to trust any source of information on network systems because most would not be suited for our game.

After gaining a general understanding of the way the multiplayer needed to be implemented, I focused more deeply on the technical concepts behind network communication. In particular, I studied sockets and their use to establish connections between machines, but also how servers and clients interact within a networked application.

### b. Technical Implementation

Once I acquired this theoretical knowledge, I began implementing my first server and client prototypes. These prototypes were not intended to represent the final version of the game but instead served as experimental tools to test network communication in a controlled environment. Their main purpose was to verify that multiple clients could connect to the same server and exchange data correctly.

During this phase, I encountered several technical difficulties that are common in multiplayer networking. One concrete example was a synchronization issue caused by blocking socket calls. In early versions of the server, when one client stopped sending data or experienced lag, the server would freeze temporarily while waiting for input. This caused all other connected clients to stop receiving updates, effectively halting the game. Resolving this issue required research into non-blocking sockets and better control of the server's update loop, which significantly improved responsiveness and stability.

As the project progressed, I realized that sending raw data directly between the server and clients was not a sustainable solution. This led me to design and implement a system that converts data into structured and formatted packets. These packets were organized in a way that clearly defined the type and content of each message, allowing both the server and clients to interpret incoming data reliably. This system made it possible to transmit a wide variety of information, including player actions, game states, and system events, in a consistent and extensible format. By standardizing the data exchange process, I significantly improved the reliability and maintainability of the networking system.

To further improve the organization of the project, I implemented an event handler responsible for managing common network-related actions. This event handler simplified basic operations such as player connection and disconnection by centralizing their logic. As a result, the code became cleaner, more modular, and easier to maintain. Using an event-driven approach also reduced redundancy and made it easier to add new features without disrupting existing functionality. This architectural improvement played an important role in making the system more scalable and adaptable to future development.

During the beginning of the second semester, most of the work related to the network connections was dedicated to making a link between the menu and the game. In order to have a working finished game that's working, I needed to make sure that if two players connected to the game, sent their information and tried to launch the game, the game would actually launch, with each player being registered as a client connected to a server hosted on one of the player's computer. This created some unplanned challenges: many network synchronisation functions were implemented to handle issues during the connection processes.

Then, I've had to link the two player's information using the events. That is when I realised that when the event handler and network system was at the first stage, I tested it on a very simple game (just a simple rectangle moving) while our game handles way more data than just position. Soon, it created a problem with the display of our graphisms on the screen: if the first player loads a room, every graphics being displayed on his screen, the other player wouldn't have the same design. And while this is just a graphism problem, it began impeding on the playability of the game since it also impacted the platforms. I had to implement new client and server events to handle the room generation data.

### c. Technical Improvements and GitHub

I discovered the importance of version control in collaborative projects. I learned how to use GitHub to manage and share my work with the rest of my group. This involved understanding how repositories function, how to commit changes properly, and how to push updates to a remote repository. I also encountered minor difficulties, such as merge conflicts and accidental overwrites, which taught me how to resolve conflicts and structure commits more carefully. Using GitHub allowed me to collaborate more efficiently with my teammates and ensured that everyone had access to the latest version of the project. It also provided a secure way to track changes and revert to previous versions, if necessary, which proved invaluable during experimentation and debugging.

## B) Room Structure

### a. Research

In parallel with the networking aspects of the project, I conducted research related to the visual structure of the game. I focused specifically on how backgrounds are displayed within game rooms. I studied different rendering techniques and considered their impact on performance and visual coherence. This research helped me understand how background elements can be layered to create a more immersive environment without overwhelming the player or the system.

I also researched how game rooms are created and managed using our Python-based game module built with Pygame. This allowed me to understand how rooms are loaded, how assets are handled, and how different gameplay areas are structured internally.

### b. Technical Implementation

The objective of this task was to automatically generate playable environments while maintaining a consistent design language. I focused on controlling the size of each room to ensure balance and coherence throughout the game. I also managed the spawning zones of platforms to guarantee that they appeared in logical and accessible positions. This was essential to prevent impossible level layouts and to maintain a working gameplay.

Another major aspect of this work involved the randomization of background layers. While randomness was used to create visual variety, strict rules were applied to ensure consistency. Each type of background layer always appeared at a specific depth, while

the exact sprite used for that layer was chosen randomly from a predefined set. This approach allowed environments to feel unique while preserving a coherent visual identity.

Throughout this entire phase, I continuously tested, refined, and adjusted the systems I developed. Each task contributed to a deeper understanding of both networking principles

and game design constraints. I also learned that failure and unusable research are a natural part of technical development, particularly in our multiplayer game. Overall, this first part of the project allowed me to strengthen my technical skills and apply theoretical knowledge to practical implementation.

During the third bimester, I've implemented a working room manager to handle room transitions. I've also implemented a different system for rooms that aren't randomly generated: we want the first rooms of every world to be 3 rooms where the players will get some lore by discussing with npc's and buy items with the different merchants we planned to add to the game. The last room of every world is also a non-randomized room that holds the boss (biggest and strongest enemy of each world). To make sure this was working I had to handle the room generation data efficiently and with consistency in-between rooms (which was more of a challenge than what I thought it would be).

# VI) Lucas's part

## A) Introduction

Ever since the beginning, my task was clear: make the player character. And so, when the time was right, I set out to make it. The first element we needed was a server, as rudimentary as it may be, just to test the multiplayer aspect and make sure both players could see the entirety of the game, and that everything was correctly updated everywhere which needed it. Pierre took care of this task.

Once the server was done, I could start working. Or rather, start coding, as I had already started to familiarise myself with PyGame, which is the main library used for this project. This step was achieved through reading the documentation to have a rough idea of pygame's implementation, watching videos explaining its intricacies as well as some basic concepts like physics with rectangles. Indeed, unlike almost all game engines, pygame does not have its own physics: we have to code it by hand. This is mainly due to the fact that pygame is merely a library inside of python, instead of a full on game engine like Unity or Godot.

## B) Implemented features

### a. Functional Player

Regarding the actual coding of the Player Class (as both characters will be instances of it), it needs to take care of everything player-related. The first thing I did was to create a square, as a temporary representation of the player so that I could focus on what the player does. Which is to say, basic movement for now.

Coding left and right movement was a simple task, yet required time due to multiple reasons. First of all, speed: When the user presses the appropriate keys, the player moves left or right, but finding the right speed at which we want or player to be moving is a process of trial and error, changing only a single value between each test, realising it is too fast or too slow, up until we find the one we like. Secondly, synchronicity: when a player wants to move, we have to make sure the other player sees them moving, or doesn't move themselves if they haven't pressed any keys. This was less a matter of coding the actual player class and more about linking it to the existing server, but it did bring its own set of challenges and time consumption.

Implementing the jump mechanic was slightly trickier and not as forward, as we need two things to jump: gradual elevation, and gradual descent. This introduces many new concepts that did not matter when considering only 1-dimensional movement, such as gravity and ground collisions. The words "gradual" are also very important: we do not want the player to immediately be shot upwards when jumping, nor do we want them to fall back down too

quickly. The constant trial and error to find this fragile balance is extremely time consuming. Another element is ground collision, as the player cannot be eternally subjected to gravity's pull, forever falling. This is where the rectangle physics has to be coded by hand, and where the previously mentioned research takes all its meaning, as I reproduced what was presented to me, making sure to understand it all in the process.

### b. Pretty Player

Once we had a manoeuvrable red or blue square, depending on the player, it was time to bring them to life through the addition of sprites, laid out in a spreadsheet for more efficiency, as well as animating these sprites and getting rid of the square and placeholder mentioned above. This started, once again, by researching on how to implement spritesheets in pygame. Indeed, I only knew the method of using a list of images to animate characters, which would not work with a spritesheet. That was until I found an implementation on the PyGame wiki, which was able to handle spritesheets with a class and a few simple associated methods. I took the time to understand it, modified an element to better suit our needs, and added it to the project.

With this complete, adding an image in place of the previously used square was not too difficult, trivial even, and there only remained to switch between multiple images, all parts of the spritesheet, to create the animation. As straightforward as it sounds, it took time to get right. The two main causes of issues were the position of the sprites in the spritesheet, and how fast they were switched out. Getting the position correct relied on player positions, dimensions and relative image placement within the sheet, as well as my understanding of gaps between each sprite. Place a changing variable in place of a constant by accident and suddenly you're left with an empty space, and have to carefully check the code to see where you were wrong. As for the other point, there were in fact no gaps between two different sprites with the current dimensions, which took me a moment to realise. The second main cause of issues, the speed of the animation, was one due to the fact that if left as is, sprites would be switched at every frame, which was way too fast, and the only way to regulate time was through the pygame.clock module, which would slow the entirety of the code down if used, despite needing only a portion of it to be. To fix this, I counted frames with one variable, and only swapped out the sprites when this variable would reach multiples of a specific value, while at the same time reducing it when possible to work with numbers as small as possible.

## C) The path to follow

### a. Combat

At this point, the bases for the player were mostly complete, and only 2 features were missing: attacks and item usage. However, before continuing blindly any further, we had to know what direction to take for the combat aspect of the game. This is when me and Amélie had a meeting to discuss it. Upon concluding that meeting, we had decided about everything, with some points left as candidates for potential tweaks in the future.

First and foremost, combat is not yet implemented and everything cited here is a work for the future. Skills, represented by a deck of cards, will play an important part. This deck of cards will represent all possible abilities the players can have, and one card in it will be chosen by the player among three random cards to be their ability for the current world. Cards not chosen will not be available to the player. Both players will have different cards, as well as different attacking styles, because we want each player to have fun in their own way. For this reason, player 1 will be a melee fighter, while player 2 will attack from a distance. Both players will have a “normal” and a “charged” attack, the charged attack will as expected take longer to complete but hit harder.

Cards will be obtained at the end of each world, after the boss: player 1 will be able to choose one among a few selected cards to add to their deck. Player 2 does not gain cards this way. Instead, both players will have to find a special npc in order to “sacrifice” one of player 1’s cards, losing it forever, but allowing player 2 to gain a card in the same way player 1 does. This forces players to think and even bargain for abilities, whilst insisting on the finality of each and every decision.

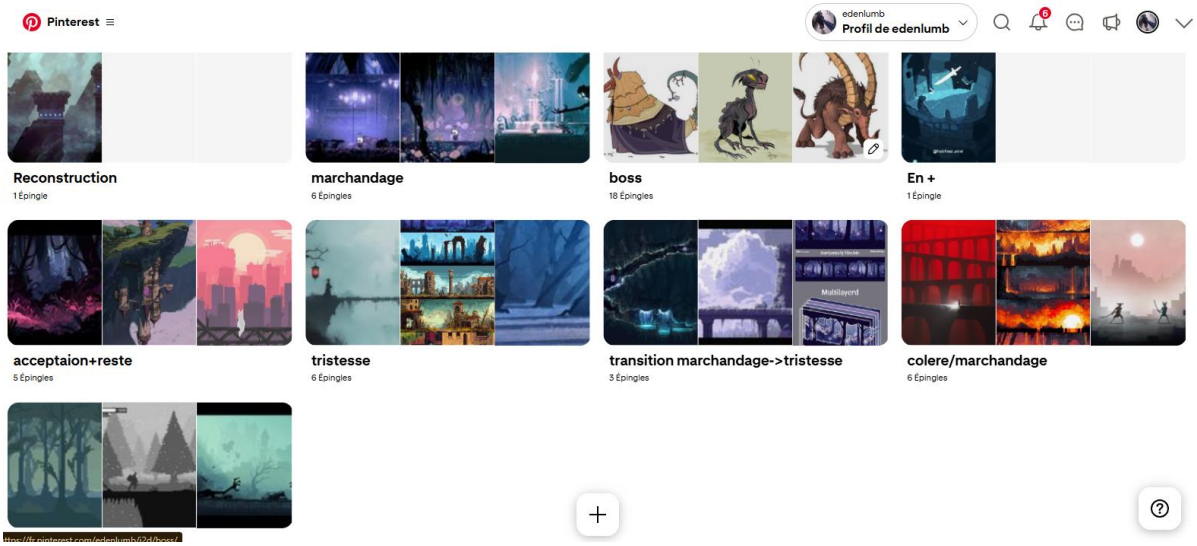
Regarding enemies, there will be at least two types: normal enemies, and strong enemies, which are slower but more tanky. All enemies will be designed according to the world they are in, to keep the experience fresh for all users.

### b. Website

My last task is to help Eden develop the website. So far, she has created a skeleton that we will use and add onto throughout development. The website also currently lacks any form of formatting, and we will be in charge of bringing it to life whilst making sure that its appearance is approved by all members of the group. This will be achieved through the use of the CSS language, which I have some familiarity with thanks to having worked with it in the past.

# Appendices :

## Appendix n°1

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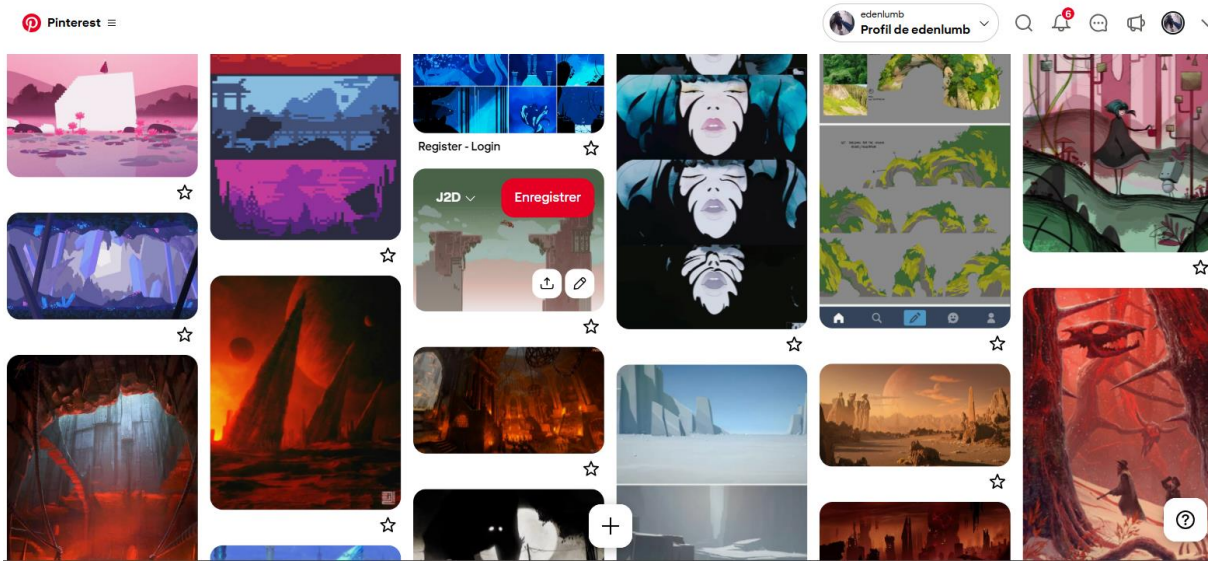
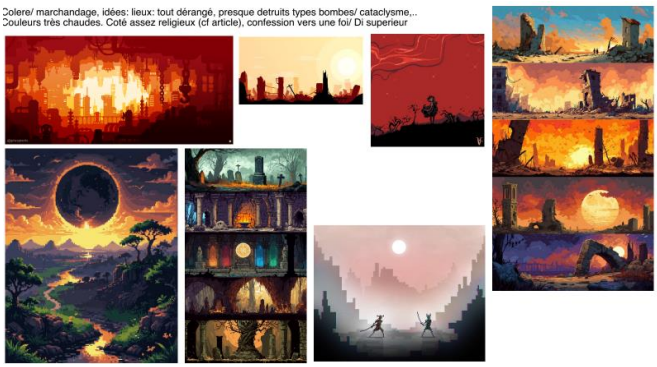



## Appendix n°3

## Appendix n°4



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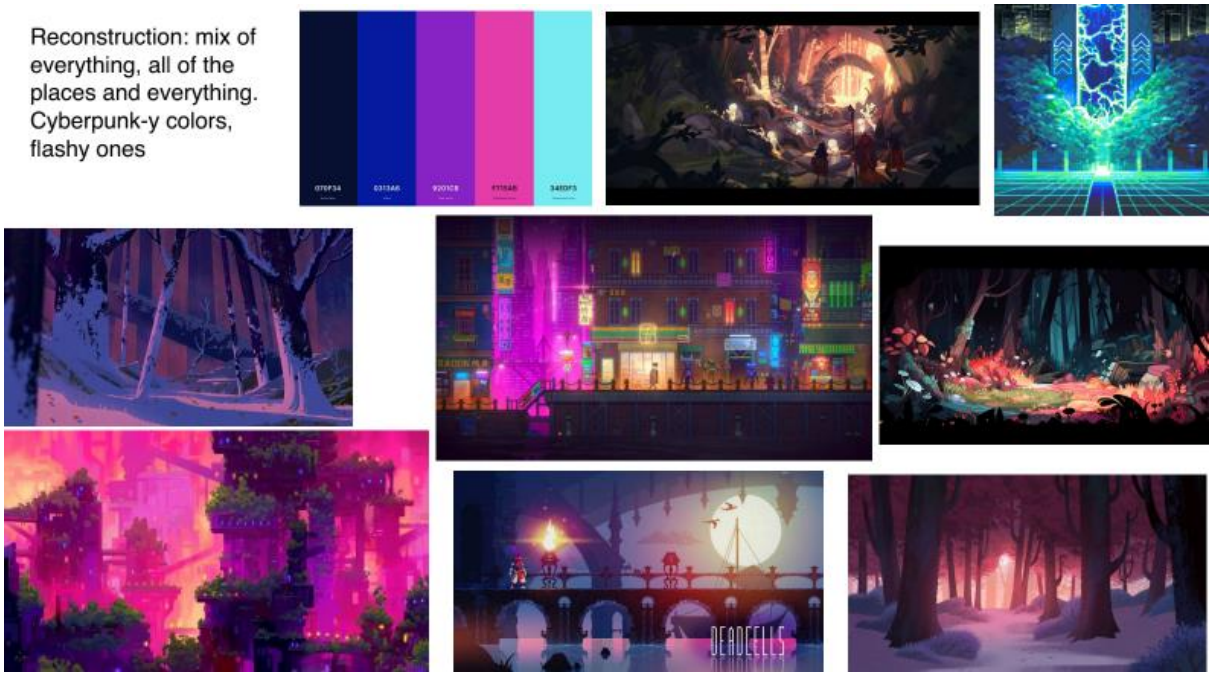


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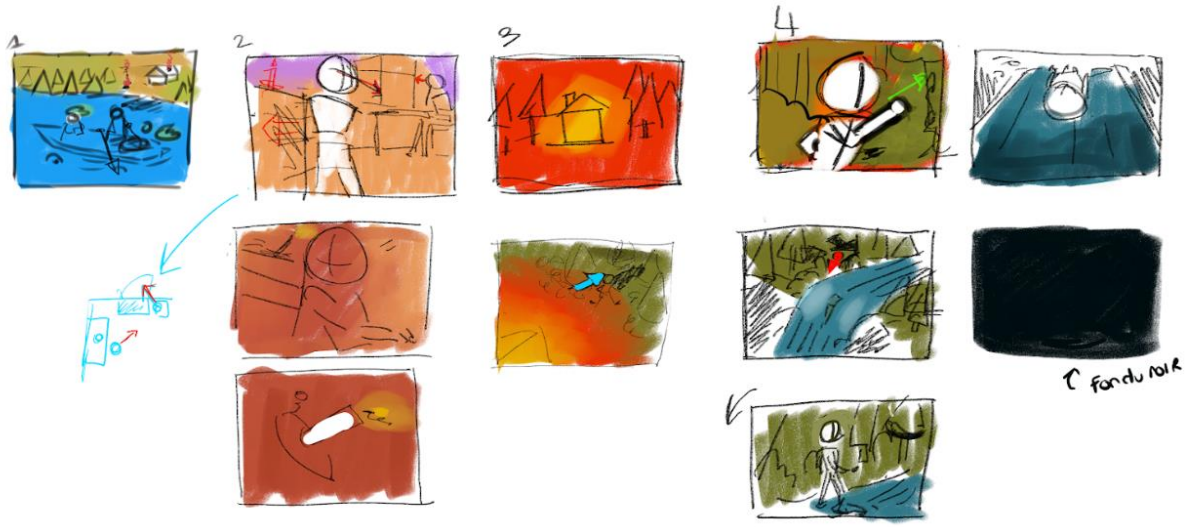


Appendix n°7

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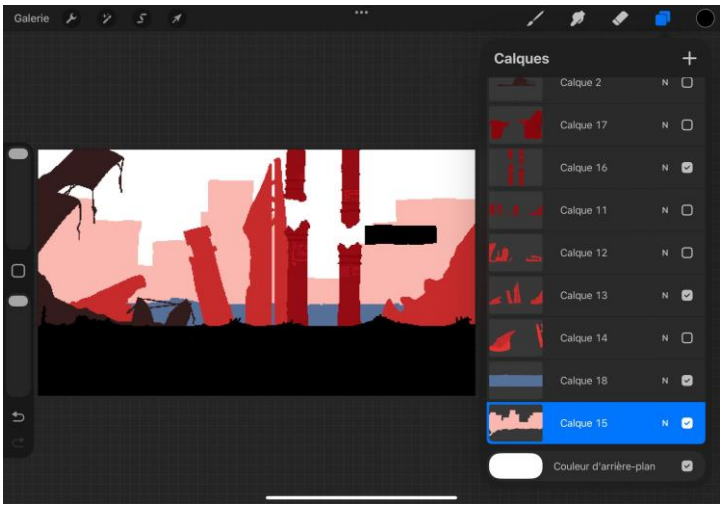
Appendix n°11




Appendix n°12

Appendix n°13

# lossless

website of lossless

our game
presentation with images
presentation of our group
all used
downloads

Appendix n°14

## GAME

### description of the game

For you to understand our game a little bit better, i will explain its story. We find ourselves brought into the life of a small family, a father and a daughter, living in a small house in the forest. After rushing to see her dad, the young girl, Belladona bumps into a candle, which becomes the turning point of the story. The entire house burns very rapidly, trapping inside the dad, who managed to get his daughter out of the place in a short time. Ordered by her father, she ran through the forest to find shelter, but encounters a new character, a lost heart. Filled with sadness, fear, and remorse of leaving her father, Belladona decides that this heart will be her companion during her travel. After some time, both of them decided to stop at a river to rest, but as they were bending over to reach the water, Belladona fell into it. Controled by the current, she could not stop, and realized that she was going to plunge into a waterfall. Without managing to carry herself out of the water, she began her downfall, passing out during her drop. When she finally wakes up, she is lying on the ground, soaked in water, the heart sitting next to her. From that point, Belladona has to make her own path to reach the end of her journey, hopefully a better place. She walks though seven worlds, the seven stages of grief as we know. For each of these worlds, she is going to have to fight her own ennemies : the villains that only want to retain her from growing, evolving, and moving on. Both characters follow the river up, and will be confronted to many tedious and bizarre adventures.

Appendix n°15

**PRESENTATION**

presentation with our images of game

Appendix n°16

**GROUP**

**description of our group, the members.**

Our group is a very diverse one. Indeed, we are five very different people, some of us are very artistic whale some others are more technical oriented, but we mobilized our ideas to create an interesting game. It is a two player game in 2D where each one has a different type of ability. It is a Roguelike game with five different levels representing the 5 known stages of grief: denial, anger, marchandise, sadness, resignation, acceptation and reconstruction. It will be represented through the 1st player character, the second player representing the accumulation of the grief itself.

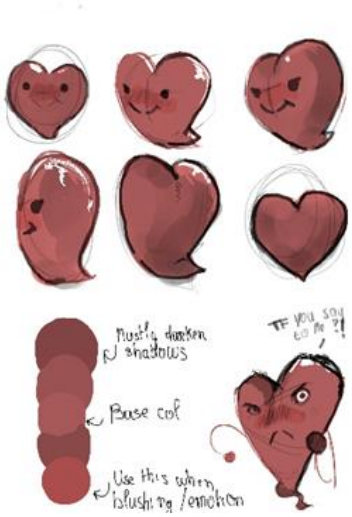
Appendix n°17

# REF

**all of the references used, books, ...**

To create our game, we used two books are references: Making games with Python and Pygame, and Pygame by Benoit Prieur

Appendix n°18

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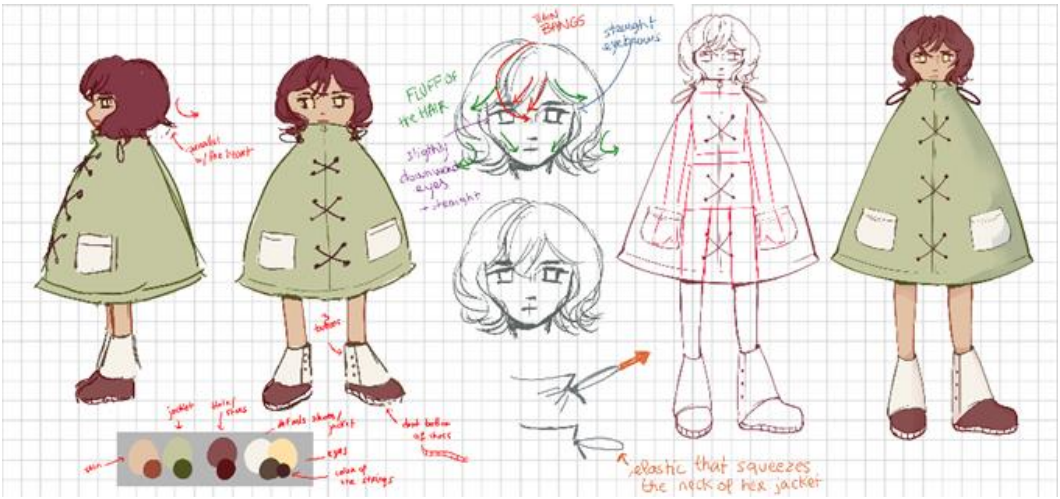

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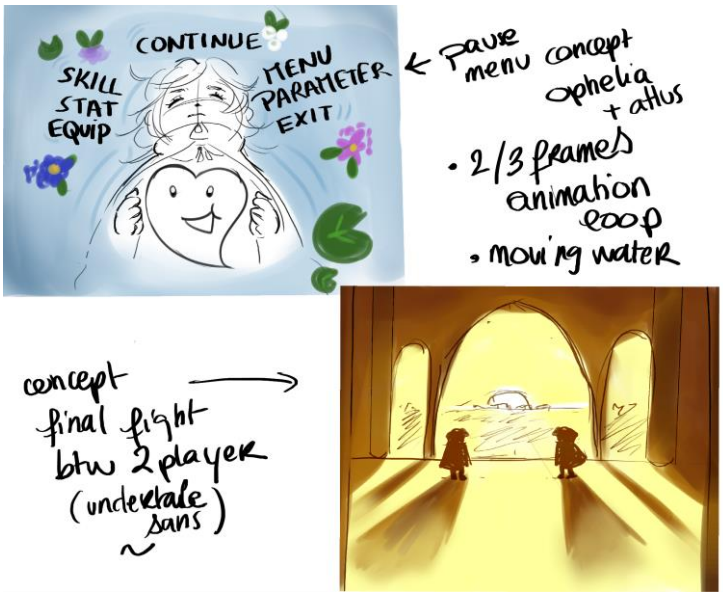


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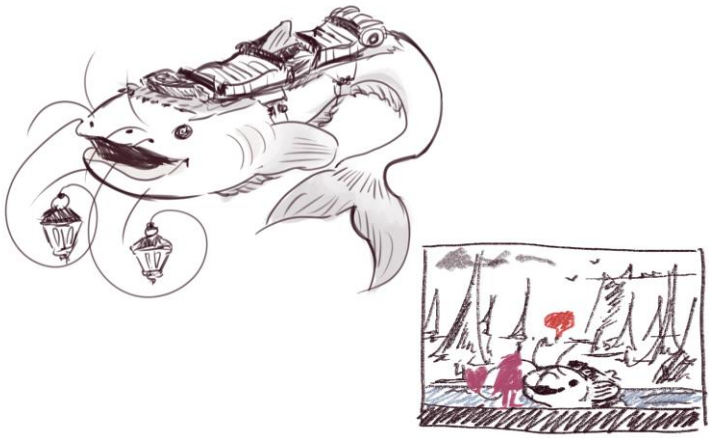

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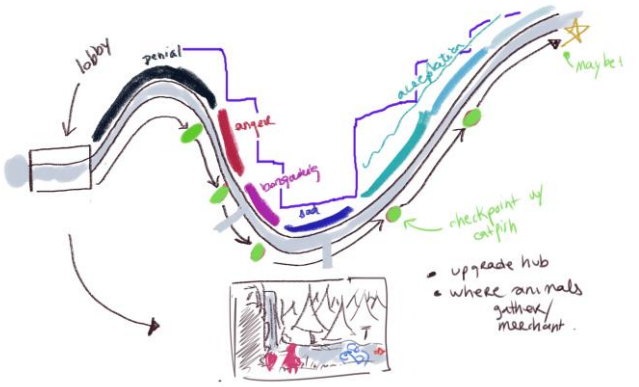


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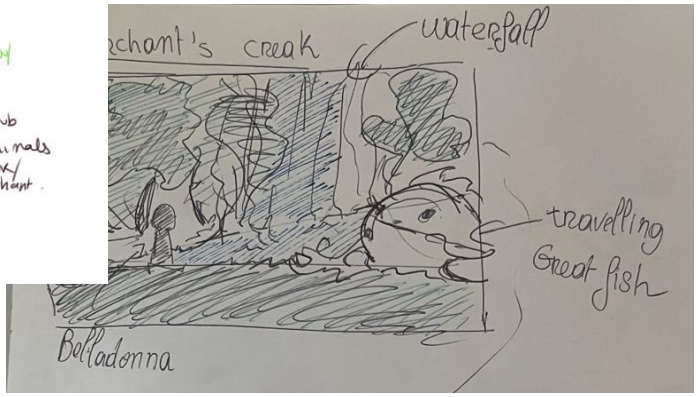


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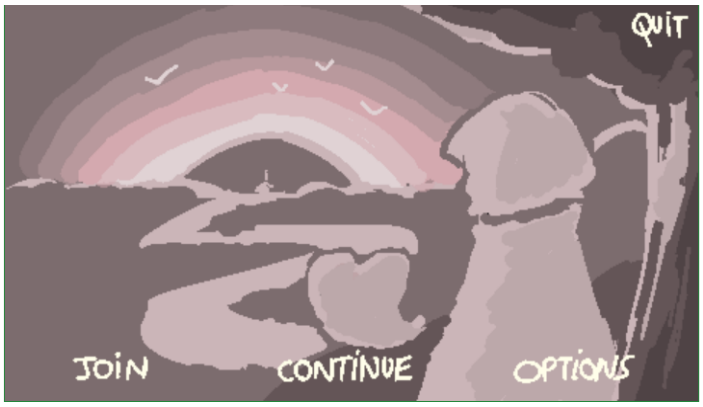


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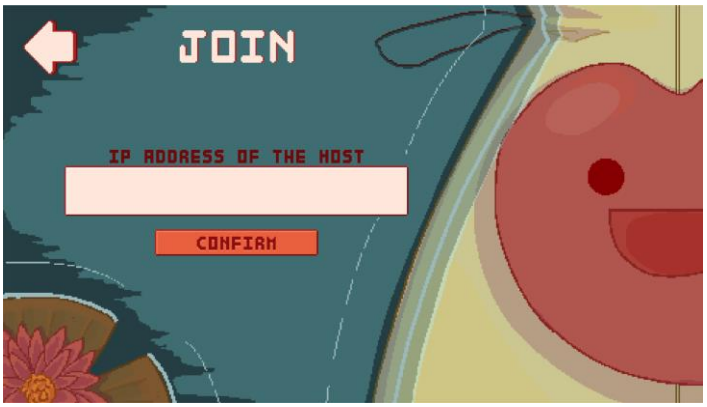


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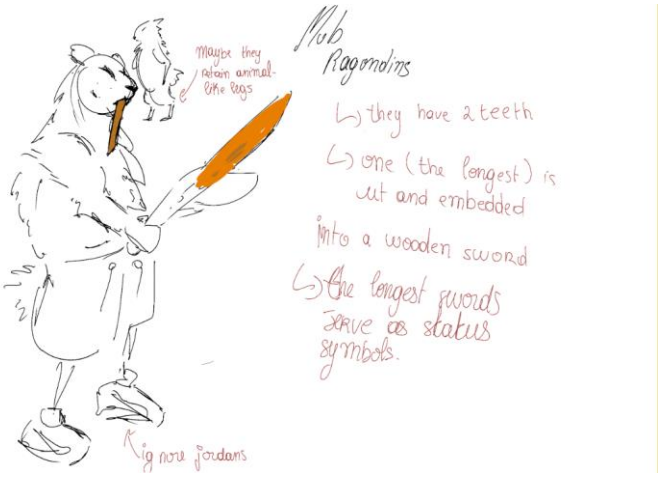
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